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Python Fundamentals for Machine Learning

Implement basic python programs for data handling, arrays, loops, functions and visualization. Learn NumPy and Pandas operations. Machine Learning Basics using Scikit-learn

- a. Create two (3*3) matrices using NumPy and print it. Perform and print the results of the following linear algebra operations
 - a. Matrix addition
 - b. Matrix subtraction
 - c. Matrix Multiplication (element-wise and dot product)
 - d. Transpose of a matrix
 - e. Determinant and inverse (if applicable)
- b. Create a Series from a list of integers representing daily temperatures (in Celsius) over a week. Assign index labels as day of the week.
 - a. Find and print the average (mean) temperature for the week.
 - b. Identify and print the maximum and minimum temperatures and their respective days.
 - c. Display the temperatures greater than a specific value.
 - d. Convert all temperatures to Fahrenheit.
 - e. Print the days had temperatures above the average.
- c. Create a data frame with details of 10 students and columns as RollNumber, Name, Gender, Marks1, Marks2, Marks3.
 - a. Create a new column with total marks
 - b. Find the lowest marks in Marks1
 - c. Find the Highest marks in Marks2
 - d. Find the average marks in Marks3
 - e. Find student name with highest average
 - f. Find how many students failed in Marks2 (<40).

Question A:

```
import numpy as np

class Matrix:
    def matrix_add(matrix1,matrix2):
        return np.add(matrix1,matrix2)

    def matrix_subtract(matrix1,matrix2):
        return np.subtract(matrix1,matrix2)

    def matrix_multiply(matrix1,matrix2):
        return np.multiply(matrix1,matrix2)

    def matrix_division(matrix1,matrix2):
        return np.divide(matrix1,matrix2)

    def matrix_dot(matrix1,matrix2):
        return np.dot(matrix1,matrix2)

    def matrix_transpose(matrix):
        return np.transpose(matrix)

matrix1 = np.array([
    [1, 2, 3],
    [0, 1, 4],
    [5, 6, 0]
])

matrix2 = np.array([
    [7, 8, 1],
    [2, 0, 3],
    [4, 5, 2]
])

print("Matrix-A:",matrix1)
print("Matrix-B:",matrix2)

print("\n````Matrix opearations````")
print(f"Matrix Addition:\n{matrix1}+{matrix2}:\n",Matrix.matrix_add(matrix1,matrix2))
print("\n")
print(f"Matrix Subtraction:\n{matrix1}-{matrix2}:\n",Matrix.matrix_subtract(matrix1,matrix2))
print("\n")
print(f"Matrix Multiplication:\n{matrix1}X{matrix2}:\n",Matrix.matrix_multiply(matrix1,matrix2))
print("\n")
print(f"Matrix Division:\n{matrix1}/{matrix2}:\n",Matrix.matrix_division(matrix1,matrix2))
print("\n")
print(f"Transpose of matrix1: {matrix1} is",Matrix.matrix_transpose(matrix1))
print("\n")
```

```

print(f"Transpose of matrix2: {matrix2} is",Matrix.matrix_transpose(matrix2))
print("\n")
print(f"Matrix Division:\n{matrix1} . {matrix2}:\n",Matrix.matrix_dot(matrix1,matrix2))
print("\n")

```

Output:

```

MINGW64:/d/Lab-1
(.venv)
user@Dhyana MINGW64 /d/Lab-1
$ python question_a.py
Matrix-A: [[2 1 3]
[1 4 1]
[3 2 5]]
Matrix-B: [[3 5 2]
[2 1 4]
[1 3 2]]

'''Matrix opearations'''
Matrix Addition:
[[2 1 3]
[1 4 1]
[3 2 5]]+[[3 5 2]
[2 1 4]
[1 3 2]]:
[[5 6 5]
[3 5 5]
[4 5 7]]

Matrix Subtraction:
[[2 1 3]
[1 4 1]
[3 2 5]]-[[3 5 2]
[2 1 4]
[1 3 2]]:
[[-1 -4 1]
[-1 3 -3]
[2 -1 3]]

Matrix Multiplication:
[[2 1 3]
[1 4 1]
[3 2 5]]x[[3 5 2]
[2 1 4]
[1 3 2]]:
[[ 6  5  6]
[ 2  4  4]
[ 3  6 10]]

Matrix Division:
[[2 1 3]
[1 4 1]
[3 2 5]]/[[3 5 2]
[2 1 4]
[1 3 2]]:
[[0.66666667 0.2 1.5]
[0.5 4. 0.25]
[3. 0.66666667 2.5]]

Transpose of matrix1:[[2 1 3]
[1 4 1]
[3 2 5]] is [[2 1 3]
[1 4 2]
[3 1 5]]

Transpose of matrix2:[[3 5 2]
[2 1 4]
[1 3 2]] is [[3 2 1]
[5 1 3]
[2 4 2]]

Matrix Division:
[[2 1 3]
[1 4 1]
[3 2 5]].[[3 5 2]
[2 1 4]
[1 3 2]]:
[[11 20 14]
[12 12 20]
[18 32 24]]

(.venv)
user@Dhyana MINGW64 /d/Lab-1
$

```

Question B:

```
data = {"Monday":22, "Tuesday":34, "Wednesday":20, "Thursday":31,
        "Friday":29,"Saturday":25, "Sunday":27
}

print("Find and print the average (mean) temperature for the week.\n")
average = sum(data.values())/len(data)
print("Average:",average,"C")

print("Identify and print the maximum and minimum temperatures and their respective days.\n")
max_pair = max(data.items(), key=lambda item: item[1])
min_pair = min(data.items(), key=lambda item: item[1])
print("Maximum Temperature:",max_pair)
print("Minimum Temperature:",min_pair)

print("Display the temperatures greater than a specific value.\n")
temperature = float(input("Enter the value to check above: "))
for day, temp in data.items():
    if temp >= temperature:
        print(f'{day}: {temp} °C')

print("Convert all temperatures to Fahrenheit.\n")
for day,temp in data.items():
    data[day] = (temp*9/5)+32
print(data)

print("Print the days had temperatures above the average\n")
for day,temp in data.items():
    if temp>(average*9/5)+32:
        print(f'{day}: {temp}')
```

Output:

```
MINGW64:/d/Lab-1
(.venv)
user@Dhyana MINGW64 /d/Lab-1
$ python question_b.py
Find and print the average (mean) temperature for the week..

Average: 26.857142857142858
Identify and print the maximum and minimum temperatures and their respective days.
Maximum Temperature: ('Tuesday', 34)
Minimum Temperature: ('Wednesday', 20)
Display the temperatures greater than a specific value.
Enter the value to check above: 26
Tuesday: 34°C
Thursday: 31°C
Friday: 29°C
Sunday: 27°C
Convert all temperatures to Fahrenheit.
{'Monday': 71.6, 'Tuesday': 93.2, 'Wednesday': 68.0, 'Thursday': 87.8, 'Friday': 84.2, 'Saturday': 77.0, 'Sunday': 80.6}
Print the days had temperatures above the average
Tuesday:93.2
Thursday:87.8
Friday:84.2
Sunday:80.6
(.venv)
user@Dhyana MINGW64 /d/Lab-1
```

Question C:

```
import pandas as pd

student_data = pd.DataFrame({
    'Roll Number': [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    'Name': ['Deepika Padukone', 'Shah Rukh Khan', 'Amitabh Bachchan', 'Ranbir Kapoor',
            'Alia Bhatt', 'Hrithik Roshan', 'Katrina Kaif', 'Priyanka Chopra', 'Prabhas', 'Vijay'],
    'Gender': ['F', 'M', 'M', 'M', 'F', 'M', 'F', 'F', 'M', 'M'],
    'Marks1': [85, 72, 90, 68, 95, 78, 88, 62, 8, 75],
    'Marks2': [88, 75, 12, 70, 40, 82, 85, 65, 79, 80],
    'Marks3': [9, 78, 8, 65, 98, 80, 91, 70, 5, 82]
})

df = student_data
print(df)
print("\n")

print("Create a new column with total marks")
df["Total Marks"] = df["Marks1"]+df["Marks2"]+df["Marks3"]
print(df)
print("\n")

print("Find the lowest marks in Marks1:",df["Marks1"].min())
print("\n")

print("Find the Highest marks in Marks2:",df["Marks2"].max())
print("\n")

print("Find the average marks in Marks3",df["Marks3"].mean())
print("\n")

print("Find student name with highest average:")
df["Average Marks"] = df["Total Marks"]/3
top = df["Average Marks"].idxmax()
print(df.loc[top, "Name"])
print(df.loc[top, "Average Marks"])
print("\n")

print("Find how many students failed in Marks2 (<40)")
count = 0
for mark in df["Marks2"]:
    if mark < 40:
        count += 1
print("Number of students failed:",count)
print("\n")
```

Output:

```
MINGW64:/d/Lab-1
(.venv)
user@Dhyana MINGW64 /d/Lab-1
$ python question_c.py
```

Roll Number	Name	Gender	Marks1	Marks2	Marks3
101	Deepika Padukone	F	85	88	9
102	Shah Rukh Khan	M	72	75	78
103	Amitabh Bachchan	M	90	12	8
104	Ranbir Kapoor	M	68	70	65
105	Alia Bhatt	F	95	40	98
106	Hrithik Roshan	M	78	82	80
107	Katrina Kaif	F	88	85	91
108	Priyanka Chopra	F	62	65	70
109	Prabhas	M	8	79	5
110	Vijay	M	75	80	82

```
Create a new column with total marks
Roll Number      Name      Gender      Marks1      Marks2      Marks3      Total Marks
101      Deepika Padukone      F      85      88      9      182
102      Shah Rukh Khan      M      72      75      78      225
103      Amitabh Bachchan      M      90      12      8      110
104      Ranbir Kapoor      M      68      70      65      203
105      Alia Bhatt      F      95      40      98      233
106      Hrithik Roshan      M      78      82      80      240
107      Katrina Kaif      F      88      85      91      264
108      Priyanka Chopra      F      62      65      70      197
109      Prabhas      M      8      79      5      92
110      Vijay      M      75      80      82      237

Find the lowest marks in Marks1: 8

Find the Highest marks in Marks2: 88

Find the average marks in Marks3 58.6

Find student name with highest average:
Katrina Kaif
88.0

Find how many students failed in Marks2 (<40)
Number of students failed: 1

(.venv)
user@Dhyana MINGW64 /d/Lab-1
$
```

